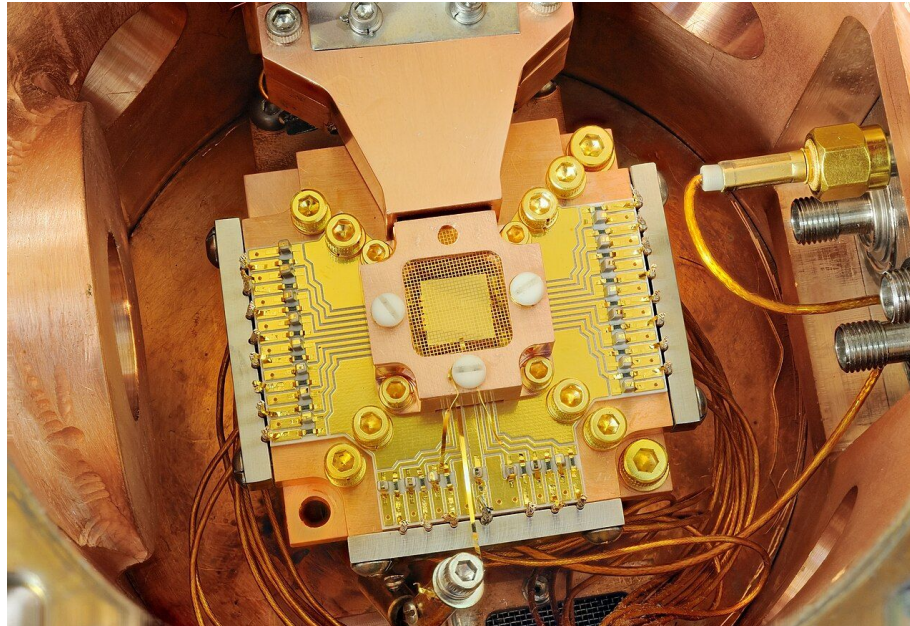


Post quantum tales from the trenches

TechWorks Summit - August 2026



Quantum frontiers may be closer than they appear

Mar 25, 2026

1 min read

We're setting a timeline for post-quantum cryptography migration to 2029.

<https://blog.google/innovation-and-ai/technology/safety-security/cryptography-migration-timeline/>



Hi, I'm Chris

@cpswan

<https://chris.swanz.net>



Agenda

- Crypto Agility first
- System libraries only on recent distributions
- Options for native code
 - Ensuring compatibility
- Hybrid mechanisms haven't yet standardised



Crypto Agility

Likely to find LOTS of hard coded things
(and underlying assumptions)

- Algorithms
- Key types
- Key formats



OpenSSL 3.5.0 with
ML-KEM and ML-DSA
released in April 2025

Mainstream Linux Distributions carrying OpenSSL 3.5.0+

- Debian 13 (Trixie)
- Fedora 42
- RHEL 10
- Ubuntu 26.04



Our native code fallback

pqcrypto 0.4.0 

Published 46 days ago

SDK

DART

FLUTTER

PLATFORM

ANDROID

IOS

LINUX

MACOS

WEB

WINDOWS

 11

Readme

Changelog

Example

Installing

Versions

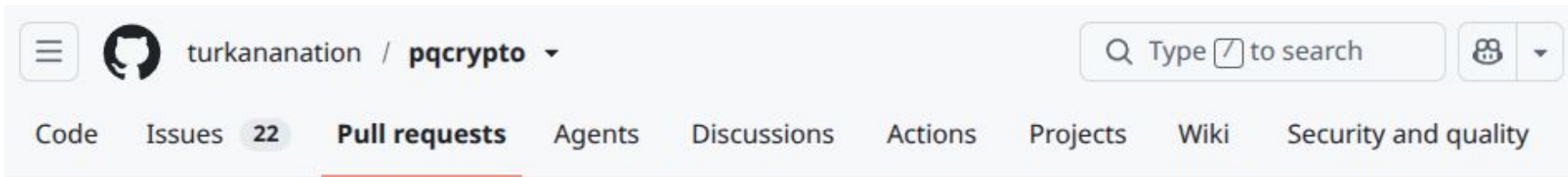
Scores

pqcrypto: Pure Dart Post-Quantum Cryptography

<https://pub.dev/packages/pqcrypto>



Getting compatibility fixed



feat: interoperability with OpenSSL #1

Merged

turkananation merged 6 commits into `turkananation:main` from `JeremyTubongbanua:main` on Jun 3

<https://github.com/turkananation/pqcrypto/pull/1>



The need for hybrid

Reasons to be wary of PQ algorithms

Single-Core PC Breaks Post-Quantum Encryption Candidate Algorithm in One Hour

News By Francisco Pires | Last updated 3 August 2022

Defeating next-gen quantum cryptography with pure math.

<https://www.tomshardware.com/news/single-core-pc-breaks-post-quantum-encryption-candidate-algorithm-in-one-hour>



NIST Post-Quantum Cryptography: SIKE's Standardization Failure

Cryptomathic : modified on 09. August 2022

Crypto-Agility

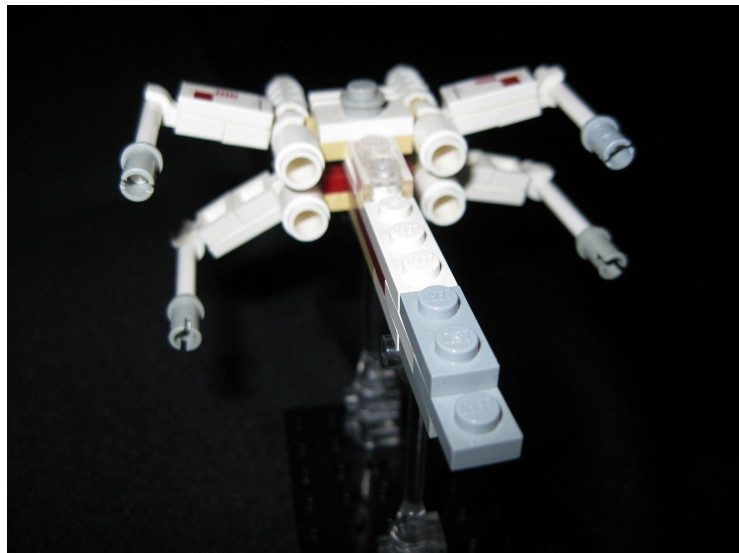
Quantum Computing

NIST

<https://www.cryptomathic.com/blog/nist-post-quantum-cryptography-standardization-sike-bites-the-dust>

Classical first or PQ?

X-Wing



Micro X-wing MOD (2) by [origamiguyl971](#) CC-BY2.0

Classical: X25519

Post-Quantum: ML-KEM-768

Combiner: SHA3-256

X-Wing binds the specific outputs of X25519 and ML-KEM-768 using SHA3-256 without needing to re-hash the large ML-KEM ciphertext, reducing compute overhead while retaining provable IND-CCA security.



<https://datatracker.ietf.org/doc/draft-connolly-cfrg-xwing-kem/>

Review

- Plan for change, and do crypto agility as a foundation for post quantum
- System libraries aren't anywhere near ubiquitous
- Native libraries are immature and may suffer compatibility issues
- X-Wing describes a well thought through hybrid mechanisms but is still an IETF draft



Thanks for your time

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Questions?